Baker's yeast β-glucan supplementation increases monocytes and cytokines post-exercise: implications for infection risk?

Strenuous aerobic exercise is known to weaken the immune system, and while many nutritional supplements have been proposed to boost post-exercise immunity, few are known to be effective. The purpose of the present study was to evaluate whether 10 d of supplementation with a defined source of baker's yeast β-glucan (BG, Wellmune WGP®) could minimise post-exercise immunosuppression. Recreationally active men and women (n 60) completed two 10 d trial conditions using a cross-over design with a 7 d washout period: placebo (rice flour) and baker's yeast BG (250 mg/d of β-1,3/1,6-glucans derived from Saccharomyces cerevisiae) before a bout of cycling (49 ± 6 min) in a hot (38 ± 2°C), humid (45 ± 2 % relative humidity) environment. Blood was collected at baseline (before supplement), pre- (PRE), post- (POST) and 2 h (2H) post-exercise. Total and subset monocyte concentration was measured by four-colour flow cytometry. Plasma cytokine levels and lipopolysaccharide (LPS)-stimulated cytokine production were measured using separate multiplex assays. Total (CD14■) and pro-inflammatory monocyte concentrations (CD14■/CD16■) were significantly greater at POST and 2H ($P<0{\cdot}05$) with BG supplementation. BG supplementation boosted LPS-stimulated production of IL-2, IL-4, IL-5 and interferon-γ (IFN-γ) at PRE and POST ($P<0{\cdot}05$). Plasma IL-4, IL-5 and IFN-γ concentrations were greater at 2H following BG supplementation. It appears that 10 d of supplementation with BG increased the potential of blood leucocytes for the production of IL-2, IL-4, IL-5 and IFN-γ. The key findings of the present study demonstrate that BG may have potential to alter immunity following a strenuous exercise session.